

GOA PANAJI, India

WMO: 431920

Lat: 15.4922N

Lon: 73.8252E

Elev: 58

StdP: 100.63

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	20.3	12.4	9.0	27.9	13.9	10.0	27.3	5.5	28.1	5.0	28.8	1.6	90	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.1	34.2	25.9	33.7	26.0	33.3	25.8	28.2	32.1	27.9	31.7	27.6	31.2	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.2	30.2	27.0	22.8	30.1	26.7	22.4	29.7	91.4	32.4	89.9	31.8	88.5	31.4	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.1	5.2	DB	17.8	36.0	1.4	0.9	16.8	36.6	16.0	37.1	15.3	37.6	14.2	38.3	(4)
			WB	14.7	29.0	0.9	0.6	14.0	29.4	13.5	29.8	13.0	30.2	12.3	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	26.3	26.8	28.2	29.6	30.1	27.9	27.0	27.2	28.1	28.0	27.0		
		DBStd	1.62	1.19	1.21	1.19	0.96	1.17	1.73	1.25	1.02	0.95	1.12	1.11	1.22	(7)
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0		
		CDD10.0	6483	506	469	563	588	624	536	528	517	560	539	526		
		CDD18.3	3441	248	236	304	338	366	286	269	268	267	302	289	267	(11)
		CDH23.3	36521	2145	2220	3417	4393	4913	3176	2612	2536	2548	3157	2974	2431	(12)
		CDH26.7	13957	873	910	1406	2082	2487	1112	585	513	633	1123	1223	1012	(13)
(14)	Wind	WSAvg	1.9	1.5	1.6	1.7	1.9	2.2	2.7	3.2	2.5	1.7	1.4	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	2846	1	0	2	5	89	849	915	563	267	128	29	13	(15)
		PrecMax	4315	10	3	43	45	729	1586	1526	1193	764	391	191	216	(16)
		PrecMin	1723	0	0	0	0	0	111	303	98	23	1	0	0	(17)
		PrecStd	503	2	1	8	8	137	272	285	238	163	105	45	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	34.0	34.3	34.4	34.4	34.8	34.1	31.2	30.8	31.7	34.1	34.3	34.1		
		MCWB	22.1	21.8	24.1	26.7	27.6	27.6	27.4	26.9	26.9	25.6	24.0	22.4		
		DB	32.7	33.1	33.1	33.7	34.2	33.0	30.5	30.2	31.0	33.1	33.4	33.0		
		MCWB	22.2	22.7	25.5	27.0	27.5	27.2	26.8	26.6	25.9	24.2	22.9	22.9		
		DB	31.8	32.1	32.4	33.2	33.7	31.9	29.9	29.7	30.3	32.2	32.7	32.2		
		MCWB	22.2	23.2	25.7	26.8	27.3	27.4	27.0	26.6	26.4	26.1	24.3	22.8		
		DB	30.9	31.2	31.8	32.7	33.2	30.8	29.2	29.1	29.7	31.3	31.9	31.4		
		MCWB	22.2	23.5	25.6	26.6	27.0	27.2	26.7	26.5	26.2	26.2	24.4	22.6		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	25.7	26.6	27.7	28.5	28.7	28.6	28.0	27.7	27.7	28.2	27.3	26.3		
		MCDB	30.1	30.8	32.1	33.1	33.1	31.9	30.3	29.7	30.4	31.7	31.3	30.9		
		WB	24.7	25.9	27.2	28.0	28.2	28.1	27.5	27.2	27.2	27.6	26.6	25.5		
		MCDB	29.2	29.9	31.4	32.4	32.6	31.3	29.7	29.4	29.8	31.0	30.6	29.8		
		WB	24.1	25.2	26.8	27.6	27.9	27.7	27.2	26.9	26.9	27.1	26.1	24.9		
		MCDB	28.9	29.3	30.7	31.8	32.2	30.7	29.3	29.0	29.4	30.4	30.0	29.0		
		WB	23.4	24.5	26.4	27.2	27.6	27.3	26.8	26.6	26.5	26.7	25.7	24.3		
		MCDB	28.3	28.8	30.2	31.3	31.8	30.1	28.8	28.5	28.9	29.8	29.4	28.4		
(35)	Mean Daily Temperature Range	MDBR	10.5	9.8	7.9	6.7	6.1	4.5	3.6	3.9	5.0	6.8	9.0	10.3		
		5% DB	11.4	10.8	8.2	6.8	6.5	6.0	4.4	4.6	5.8	8.2	9.8	11.0		
		MCWBR	4.2	4.2	3.3	2.8	2.2	2.4	2.3	2.3	2.5	3.1	3.4	4.0		
		5% WB	9.9	8.9	7.1	6.3	6.2	5.3	4.2	4.3	5.2	6.7	7.9	8.8		
		MCWBR	4.2	4.1	3.0	2.7	2.6	2.7	2.4	2.5	2.6	3.2	3.1	3.5		
(40)	Clear-Sky Solar Irradiance	taub	0.459	0.472	0.553	0.607	0.582	0.610	0.659	0.601	0.525	0.575	0.515	0.471		
		taud	2.181	2.158	1.969	1.873	1.942	1.892	1.770	1.905	2.120	1.950	2.083	2.160		
		Ebn at Noon	829	838	780	737	745	716	684	732	792	738	770	804		
		Edh at Noon	143	152	189	208	191	199	225	199	160	185	156	143		
(44)	All-Sky Solar Radiation	RadAvg	5.36	6.13	6.64	6.88	6.45	4.16	3.80	4.26	4.91	5.12	5.10	5.07		
		RadStd	0.20	0.17	0.24	0.20	0.37	0.44	0.57	0.36	0.45	0.30	0.28	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.14	N/A	+0.61	+0.34	+0.39	+0.33	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+0.15	N/A	+0.86	+0.37	+0.36	+0.34	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page